

# COMPLETE CLASSIFICATION OF PRIMITIVE INTEGER SOLUTIONS TO RAMANUJAN-SATO SERIES UNDER THE GROTHENDIECK PERIOD CONJECTURE

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ABSTRACT.

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## 1. INTRODUCTION

For  $\ell = 1, 2, 3, 4$ , let

$$\begin{aligned} T_1(m) &= \binom{6m}{3m} \binom{3m}{m} \binom{2m}{m} = \frac{(6m)!}{(3m)!(m!)^3}, \\ T_2(m) &= \binom{4m}{2m} \binom{2m}{m}^2 = \frac{(4m)!}{(m!)^4}, \\ T_3(m) &= \binom{3m}{m} \binom{2m}{m}^2, \\ T_4(m) &= \binom{2m}{m}^3. \end{aligned}$$

Define

$$F_\ell(X) = \sum_{m \geq 0} T_\ell(m) X^m, \quad \Theta = X \frac{d}{dX},$$

and

$$S_\ell(A, B, C) = \sum_{m \geq 0} T_\ell(m) \frac{A + Bm}{C^m}, \quad A, B, C \in \mathbb{Z}, \quad C \neq 0, \quad \gcd(A, B) = 1.$$

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Within the radius of convergence,

$$S_\ell(A, B, C) = AF_\ell(C^{-1}) + B\Theta F_\ell(C^{-1}).$$

**Period torsor hypothesis.** Let  $E/\overline{\mathbb{Q}}$  be an elliptic curve,

$$V = H^1(E), \quad M_E = \mathrm{Sym}^2 V \oplus \mathbb{Q}(-1).$$

Denote by  $\mathfrak{T}_E = \langle M_E \rangle^\otimes$  the pure André motive Tannakian category generated by  $M_E$ , and by  $\Omega_E^{\mathrm{mot}}$  its motivic period torsor.

(GPC<sub>tor</sub>). The complex point given by the comparison isomorphism is Zariski dense in  $\Omega_E^{\mathrm{mot}}$ . Equivalently, any polynomial period relation with coefficients in  $\overline{\mathbb{Q}}$  that holds at the comparison isomorphism holds identically on the whole motivic period torsor.

This is the full period conjecture as stated by Bost–Charles, “the comparison isomorphism is dense in the motivic period torsor”, not merely the usual fixed codimension algebraic cycle version. Below we only use its consequence for linear relations (linear relations among matrix coefficients).

**Theorem 1.1** (Main theorem). *Fix  $\ell \in \{1, 2, 3, 4\}$ . Assume that (GPC<sub>tor</sub>) holds for every elliptic curve  $E/\overline{\mathbb{Q}}$  that appears in the proof. Then there are only finitely many primitive integer triples  $(A, B, C)$  satisfying*

$$S_\ell(A, B, C) = \frac{\alpha}{\pi}, \quad \alpha \in \overline{\mathbb{Q}},$$

*and such that the series converges. For each fixed  $C$ , the primitive integer pair  $(A, B)$  has at most one projective slope; hence at most one pair  $(A, B)$  up to sign  $(A, B)$  and  $(-A, -B)$ . This statement includes the case  $\alpha = 0$ . The proof does not use any upper bound on  $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ , so adding an arbitrary fixed degree condition  $[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq d$  does not change the conclusion.*

In addition, we classify all the level 1-4 Ramanujan-Sato series, see Section 7. Take level 2a as an example, which case all of the solutions are generated by the above 12 primitive solutions: the positive part

| $n$ | $C$         | $A$  | $B$   | $\alpha$                 |
|-----|-------------|------|-------|--------------------------|
| 4   | 648         | 1    | 7     | $\frac{9}{2}$            |
| 6   | 2304        | 1    | 8     | $2\sqrt{3}$              |
| 10  | 20736       | 1    | 10    | $\frac{9\sqrt{2}}{4}$    |
| 18  | 614656      | 3    | 40    | $\frac{49\sqrt{3}}{9}$   |
| 22  | 2509056     | 19   | 280   | $18\sqrt{11}$            |
| 58  | 24591257856 | 1103 | 26390 | $\frac{9801\sqrt{2}}{4}$ |

and the negative part

| $N$ | $C$        | $A$  | $B$   | $\alpha$                |
|-----|------------|------|-------|-------------------------|
| 5   | -1024      | 3    | 20    | 8                       |
| 7   | -3969      | 8    | 65    | $9\sqrt{7}$             |
| 9   | -12288     | 3    | 28    | $\frac{16\sqrt{3}}{3}$  |
| 13  | -82944     | 23   | 260   | $\frac{3}{72}$          |
| 25  | -6635520   | 41   | 644   | $\frac{288\sqrt{5}}{5}$ |
| 37  | -199148544 | 1123 | 21460 | 3528                    |

Take  $n = 58$  gives Ramanujan's famous formula

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{m=0}^{\infty} \frac{(4m)!}{(m!)^4} \frac{1103 + 26390m}{396^{4m}}.$$

## 2. MODULAR PARAMETERIZATION

**Lemma 2.1.** *Let  $(a_1, a_2, a_3, a_4) = (\frac{1}{6}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2})$ , and integers  $(\kappa_1, \kappa_2, \kappa_3, \kappa_4) = (1728, 256, 108, 64)$ , then*

$$\mathfrak{F}_\ell(X) = {}_3F_2\left(a_\ell, \frac{1}{2}, 1 - a_\ell; 1, 1; \kappa_\ell X\right)$$

*Proof.*  $\Gamma(nu)$  can be written as products of  $\Gamma(u+k/n)$  use the Gauss product formula. We shall only give a proof for  $\ell = 2$  and by same reasons the lemma holds for other  $\ell$ . Set  $n = 4$  in the product formula implies

$$\Gamma(4u) = \frac{4^{4u-\frac{1}{2}}}{(2\pi)^{3/2}} \Gamma(u) \Gamma\left(u + \frac{1}{4}\right) \Gamma\left(u + \frac{1}{2}\right) \Gamma\left(u + \frac{3}{4}\right).$$

Now substitute  $u = m + \frac{1}{4}$  we have  $4u = 4m + 1$ . Plug in gives  $\Gamma(4m + 1) = 4^{4m+\frac{1}{2}} (2\pi)^{-3/2} \Gamma\left(m + \frac{1}{4}\right) \Gamma\left(m + \frac{1}{2}\right) \Gamma\left(m + \frac{3}{4}\right) \Gamma(m + 1)$ . Since  $\Gamma(4m + 1) = (4m)!$  and  $\Gamma(m + 1) = m!$ , while noticing that  $\Gamma(m + \alpha)/\Gamma(\alpha) = (\alpha)_m$ , we obtain

$$(4m)! = \frac{4^{4m+\frac{1}{2}}}{(2\pi)^{3/2}} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{3}{4}\right) (1)_m \left(\frac{1}{4}\right)_m \left(\frac{1}{2}\right)_m \left(\frac{3}{4}\right)_m.$$

Choose  $m = 0$  gives the constant  $\frac{4^{\frac{1}{2}}}{(2\pi)^{3/2}} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{3}{4}\right) = 1$ , and therefore

$T_m = 4^{4m} \frac{(\frac{1}{4})_m (\frac{1}{2})_m (\frac{3}{4})_m}{(1)_m^3}$ . The other three cases follow from the sextuple, triple and double multiplication formulas respectively.  $\square$

**Lemma 2.2.** *For each  $\ell \in \{1, 2, 3, 4\}$ , there exists a constant  $c_\ell > 0$  such that*

$$T_\ell(m) = c_\ell \frac{\kappa_\ell^m}{m^{3/2}} \left(1 + O\left(\frac{1}{m}\right)\right).$$

Therefore, the convergence of the series  $S_\ell(A, B, C)$  is as follows:

| Condition on $C$    | Convergence of $S_\ell(A, B, C)$                 |
|---------------------|--------------------------------------------------|
| $ C  > \kappa_\ell$ | absolute convergence,                            |
| $C = \kappa_\ell$   | convergent iff $B = 0$ ,                         |
| $C = -\kappa_\ell$  | always convergent; conditionally if $B \neq 0$ , |
| $ C  < \kappa_\ell$ | divergent.                                       |

*Proof.* By Lemma 2.1,

$$T_\ell(m) = \kappa_\ell^m \frac{\Gamma(m + a_\ell) \Gamma(m + \frac{1}{2}) \Gamma(m + 1 - a_\ell)}{\Gamma(a_\ell) \Gamma(\frac{1}{2}) \Gamma(1 - a_\ell) \Gamma(m + 1)^3}.$$

By using the Stirling formula,  $m! \sim 1/\sqrt{2\pi m} \cdot (m/e)^m$ , we have the Gamma quotient  $\Gamma(m + r)/\Gamma(m + 1) = m^{r-1} (1 + O(m^{-1}))$ , therefore the total exponent of  $m$  is  $(a_\ell - 1) + (\frac{1}{2} - 1) + (-a_\ell) = -\frac{3}{2}$ , and the constant is given by

$$c_\ell = \frac{1}{\Gamma(a_\ell) \Gamma(\frac{1}{2}) \Gamma(1 - a_\ell)} > 0.$$

When  $|C| > \kappa_\ell$ , the ratio test gives absolute convergence; when  $|C| < \kappa_\ell$ , the general term does not tend to zero. When  $C = \kappa_\ell$ , if  $B \neq 0$ , the terms have constant sign and are of order  $Bm^{-1/2}$ , hence diverge; if  $B = 0$ , they are of order  $m^{-3/2}$ , hence converge. When  $C = -\kappa_\ell$ , the terms alternate; if  $B \neq 0$ , their absolute values are of order  $m^{-1/2}$ , giving conditional convergence; if  $B = 0$ , the convergence is absolute.  $\square$

We introduce the notation  $q = e^{2\pi i \sigma}$ ,  $\sigma \in \mathfrak{H}$ . For  $\ell = 1, 2, 3, 4$ , let  $Y_\ell = X_0(\ell)^+$  be the Fricke quotient; for  $\ell = 1$  we set  $Y_1 = X(1)$ . Campbell–Cooper–Ye take a Hauptmodul  $X_\ell = q + O(q^2)$  defined over  $\mathbb{Q}$ , and a weight 2 modular form  $Z_\ell = 1 + O(q)$  satisfying

$$Z_\ell = \frac{1}{\sqrt{G_\ell(X_\ell)}} q \frac{d}{dq} \log X_\ell,$$

where

| $\ell$ | $G_\ell(X)$ | $H_\ell(X)$ |
|--------|-------------|-------------|
| 1      | $1 - 1728X$ | $240X$      |
| 2      | $1 - 256X$  | $48X$       |
| 3      | $1 - 108X$  | $24X$       |
| 4      | $1 - 64X$   | $16X$       |

**Lemma 2.3** (Modular form equals hypergeometric solution). *In a neighborhood of  $q = 0$ ,  $Z_\ell(q) = \mathfrak{F}_\ell(X_\ell(q))$ . Consequently,*

$$q \frac{dX_\ell}{dq} = X_\ell Z_\ell \sqrt{1 - \kappa_\ell X_\ell}.$$

*Proof.* Write

$$Z_\ell = \sum_{n \geq 0} U_\ell(n) X_\ell^n, \quad U_\ell(0) = 1.$$

The recurrence of Campbell–Cooper–Ye, with  $G = 1 - \kappa X$ ,  $H = hX$ , is given by

$$(n+1)^3 U(n+1) = \left(n + \frac{1}{2}\right) (h + \kappa n(n+1)) U(n).$$

where  $h_\ell = \kappa_\ell a_\ell(1 - a_\ell)$ . Therefore  $h_\ell + \kappa_\ell n(n+1) = \kappa_\ell(n + a_\ell)(n + 1 - a_\ell)$ , so

$$\frac{U_\ell(n+1)}{U_\ell(n)} = \kappa_\ell \frac{(n + a_\ell)(n + \frac{1}{2})(n + 1 - a_\ell)}{(n+1)^3}.$$

This is exactly the ratio of consecutive terms of  $T_\ell(n)$  from Lemma 2.1; since both have initial value 1, we get  $U_\ell(n) = T_\ell(n)$ .

The second formula follows directly from the definition.  $\square$

**Lemma 2.4** (Algebraic parameters give algebraic elliptic curves). *Let  $x \in \overline{\mathbb{Q}}$  be a non-cuspidal value on  $Y_\ell$ . If  $X_\ell(\sigma) = x$ , then one can choose a point*

$$\tilde{P} \in X_0(\ell)(\overline{\mathbb{Q}})$$

*lying above the corresponding point; it parametrises an elliptic curve  $E_x/\overline{\mathbb{Q}}$  together with a cyclic  $\ell$ -isogeny structure. Specifically, if  $x \in \mathbb{Q}$ , one can choose  $\tilde{P}$  such that  $[\mathbb{Q}(\tilde{P}) : \mathbb{Q}] \leq 2$  (with bound 1 when  $\ell = 1$ ).*

*Proof.*  $X_\ell$  is a coordinate on  $Y_\ell \simeq \mathbf{P}^1$  defined over  $\mathbb{Q}$ , thus an algebraic number  $x$  gives an algebraic point  $P_x \in Y_\ell$ . The natural map

$$\pi_\ell : X_0(\ell) \longrightarrow X_0(\ell)^+$$

is the quotient by the Fricke involution for  $\ell > 1$ , of degree 2; for  $\ell = 1$  it is the identity. The inverse image of an algebraic point under a finite map consists of algebraic points. A geometric non-cuspidal point is represented, in moduli terms, by an elliptic curve together with a cyclic subgroup; to avoid automorphism issues of coarse moduli spaces, one may add an auxiliary full level  $N \geq 3$ , take the universal elliptic curve on a finite étale cover, and finally forget the auxiliary level. When  $x \in \mathbb{Q}$ , the residual field degree under the finite map is at most the degree of the map, giving the stated degree bound.  $\square$

### 3. PERIOD TORSOR AND COMPLEX MULTIPLICATION

Let  $U_\ell = \mathbb{P}^1 \setminus \{0, \kappa_\ell^{-1}, \infty\}$ . By Clausen's identity,  $\mathfrak{F}_\ell(x) = \mathfrak{G}_\ell(x)^2$ , where

$$\mathfrak{G}_\ell(x) = {}_2F_1\left(\begin{matrix} a_\ell/2, (1-a_\ell)/2 \\ 1 \end{matrix}; \kappa_\ell x\right).$$

The function  $\mathfrak{G}_\ell$  satisfies the second-order equation

$$\left[\Theta^2 - \kappa_\ell x \left(\Theta + \frac{a_\ell}{2}\right) \left(\Theta + \frac{1-a_\ell}{2}\right)\right] \mathfrak{G} = 0,$$

while  $\mathfrak{F}_\ell$ 's third-order equation is its symmetric square. For  $\ell = 1, 2, 3, 4$ , this second-order local system is the elliptic Picard–Fuchs system for the corresponding arithmetic triangle group; on a fixed finite algebraic cover, it equals  $R^1\pi_*\mathbb{Q}$  for some family of elliptic curves.

**Notations.** We fix a finite algebraic cover  $U$  of the modular curve  $X_0(\ell)$  (or  $X(1)$  when  $\ell = 1$ ) on which the universal elliptic curve exists as a smooth proper morphism  $\pi : \mathcal{E} \rightarrow U$ ; for a point  $u \in U$ , the fiber  $\mathcal{E}_u$  is the elliptic curve  $E_x$  corresponding to the modular parameter  $x$ . The sheaf  $R^1\pi_*\mathbb{Q}$  denotes the first higher direct image of the constant sheaf  $\mathbb{Q}$  on  $\mathcal{E}$  in the complex analytic topology, whose fibre at  $u$  is the Betti cohomology  $H^1(\mathcal{E}_u(\mathbb{C}), \mathbb{Q})$ .

Likewise,  $R^1\pi_*\Omega_{\mathcal{E}/U}^\bullet$  is the first derived direct image of the relative algebraic de Rham complex, a vector bundle on  $U$  whose fibre is the algebraic de Rham cohomology  $H_{\text{dR}}^1(\mathcal{E}_u/\kappa(u))$ , equipped with the Gauss–Manin connection. The comparison isomorphism  $\text{comp}$  is the Grothendieck period isomorphism, which at each fibre gives a  $\mathbb{C}$ -linear identification  $H_{\text{dR}}^1(E_x/\overline{\mathbb{Q}}) \otimes_{\overline{\mathbb{Q}}} \mathbb{C} \simeq H^1(E_x(\mathbb{C}), \mathbb{C})$ , and extends naturally to symmetric powers and tensor products. For a de Rham class  $s \in \text{Sym}^2 H_{\text{dR}}^1(E_x/\overline{\mathbb{Q}})$  and a Betti homology class  $\delta = \gamma^{\otimes 2} \in \text{Sym}^2 H_1(E_x(\mathbb{C}), \mathbb{Q})$ , the pairing  $\langle \text{comp}(s), \delta \rangle$  is the Kronecker product defined by integrating  $\text{comp}(s)$  over  $\gamma \times \gamma$ ; explicitly, if  $s = \omega_1 \otimes \omega_2$ , then

$$\langle \text{comp}(s), \gamma^{\otimes 2} \rangle = \left( \int_{\gamma} \omega_1 \right) \left( \int_{\gamma} \omega_2 \right).$$

Finally,  $F^2 \text{Sym}^2 H_{\text{dR}}^1(E_x)$  denotes the second step of the Hodge filtration on the symmetric square; since  $F^1 H_{\text{dR}}^1(E_x) = H^{1,0}(E_x)$  is the space of holomorphic differentials, this subspace is  $\text{Sym}^2(F^1 H_{\text{dR}}^1(E_x))$ , consisting of classes of pure type  $(2, 0)$ .

**Lemma 3.1.** *Let  $x \in \overline{\mathbb{Q}}, x \neq 0, \kappa_\ell^{-1}$ , and choose  $E_x$  as in Lemma 2.4. There exist two element*

$$s_0, s_1 \in \text{Sym}^2 H_{\text{dR}}^1(E_x/\overline{\mathbb{Q}}), \quad 0 \neq \delta \in \text{Sym}^2 H_1(E_x(\mathbb{C}), \mathbb{Q}),$$

*s.t.*

$$\begin{aligned} \langle \text{comp}(s_0), \delta \rangle &= (2\pi i)^2 F_\ell(x), \\ \langle \text{comp}(s_1), \delta \rangle &= (2\pi i)^2 \Theta F_\ell(x). \end{aligned}$$

Moreover, on the real convergence interval  $-\kappa_\ell^{-1} \leq x < \kappa_\ell^{-1}, x \neq 0$ , the vectors  $s_0, s_1$  are linearly independent. One may also take

$$s_0 \in F^2 \text{Sym}^2 H_{\text{dR}}^1(E_x) \setminus \{0\}.$$

*Proof.* In the modular curve with full auxiliary level, denote the universal elliptic curve by  $\pi : \mathcal{E} \rightarrow U$ , and set

$$\mathcal{H}_{\text{dR}} = R^1\pi_*\Omega_{\mathcal{E}/U}^\bullet, \quad \mathcal{L} = F^1\mathcal{H}_{\text{dR}}.$$

The rational  $q$ -expansion of the weight 2 modular form  $Z_\ell$ , by the  $q$ -expansion principle, gives an algebraic section defined over  $\mathbb{Q}$ :  $\mathbf{z} \in H^0(U, \mathcal{L}^{\otimes 2}) \subset H^0(U, \text{Sym}^2 \mathcal{H}_{\text{dR}})$ . In the analytic trivialization of the Tate curve, the algebraic invariant differential is  $dt/t = 2\pi i \, dz$ . Therefore, after a suitable algebraic normalization, for a chosen local flat Betti period  $\gamma$ ,  $\langle \text{comp}(\mathbf{z}), \gamma^{\otimes 2} \rangle = (2\pi i)^2 Z_\ell$ . At  $X_\ell = x$ , set

$$D = X_\ell \frac{\partial}{\partial X_\ell}.$$

Use the Gauss–Manin connection commutes with period integration, we have  $D\langle \text{comp}(\mathbf{z}), \gamma^{\otimes 2} \rangle = \langle \text{comp}(\nabla_D \mathbf{z}), \gamma^{\otimes 2} \rangle$ , and since  $\gamma$  is a flat section. By Lemma 2.3,  $DZ_\ell = \Theta F_\ell(X_\ell)$ . Choose

$$s_0 = \mathbf{z}_x, \quad s_1 = (\nabla_D \mathbf{z})_x, \quad \delta = \gamma_x^{\otimes 2},$$

at the point  $x$  gives the two equalities.

We now prove linear independence. Locally write  $\mathbf{z} = f \omega^{\otimes 2}$  where  $\omega$

generates  $\mathcal{L}$ . On the real convergence interval, Clausen's formula gives  $\mathfrak{F}_\ell(x) = \mathfrak{G}_\ell(x)^2 > 0$  so  $f(x) \neq 0$ .

The Kodaira–Spencer map  $\rho : TU \rightarrow R^1\pi_*T_{\mathcal{E}/U}$  of the modular curve is an isomorphism at nondegenerate points; hence  $\nabla_D\omega \not\equiv 0 \pmod{\mathcal{L}}$  as long as  $D \neq 0$ . Modulo  $F^2\mathrm{Sym}^2\mathcal{H}_{\mathrm{dR}} = \mathrm{Sym}^2(F^1H_{\mathrm{dR}}^1(E_x))$  we have  $\nabla_D\mathbf{z} \equiv 2f\omega \otimes \nabla_D\omega \neq 0$ . Since  $\mathbf{z} \in F^2$ ,  $\mathbf{z}_x$  and  $(\nabla_D\mathbf{z})_x$  are linearly independent. Finally  $s_0 \in \mathcal{L}_x^{\otimes 2} = F^2\mathrm{Sym}^2H_{\mathrm{dR}}^1(E_x)$ .  $\square$

**Lemma 3.2.** *Let  $\mathfrak{T}$  be a semisimple neutral Tannakian category over an algebraically closed field of characteristic zero, and let  $G$  be its Tannaka group. For a simple object  $W$ , the matrix coefficient map*

$$W_{\mathrm{dR}} \otimes W_{\mathrm{B}}^\vee \rightarrow \mathcal{O}(G)$$

*is injective. If  $W_1, W_2$  are non-isomorphic simple objects, then their matrix coefficient subspaces in  $\mathcal{O}(G)$  intersect trivially. The same conclusion holds for a  $G$ -torsor: choose any geometric base point to identify the torsor with  $G$ .*

*Proof.* For a simple representation  $W$ , the image of the group algebra in  $\mathrm{End}(W)$  generates all of  $\mathrm{End}(W)$ ; this is the finite-dimensional form of Burnside's density theorem. If a tensor  $u \otimes \lambda$  gives a matrix coefficient  $\lambda(gu)$  that vanishes for all  $g \in G$ , then  $\lambda(Au) = 0$  for all  $A \in \mathrm{End}(W)$ , hence  $u = 0$  or  $\lambda = 0$ , and hence the matrix coefficient map is injective. The matrix coefficient spaces of different simple representations are different simple subcoalgebras of the representative function coalgebra. In a semisimple coalgebra, distinct simple subcoalgebras intersect trivially.  $\square$

**Lemma 3.3.** *If  $E/\overline{\mathbb{Q}}$  has no complex multiplication, let  $V = H^1(E)$ . Then  $M = \mathrm{Sym}^2 V$  is a simple object, and  $\mathrm{Hom}_{\mathrm{Mot}}(\mathbb{Q}(-1), M) = 0$ .*

*Proof.* No CM is equivalent to  $\mathrm{End}_{\mathrm{HS}}(V) = \mathbb{Q}$ . In that case the derived group of the Hodge group acts as  $\mathrm{SL}_2$  on  $V$ , and the three-dimensional representation  $\mathrm{Sym}^2$  is irreducible, so  $\mathrm{Sym}^2 V$  has no nontrivial Hodge substructure; a motivic subobject would give a Hodge substructure, hence  $M$  is simple. On the other hand,

$$V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V, \quad \bigwedge^2 V \simeq \mathbb{Q}(-1).$$

Since  $\mathrm{End}_{\mathrm{HS}}(V) = \mathbb{Q}$ , the Tate line in  $V \otimes V$  is unique and given by the polarisation; it lies in the alternating square  $\bigwedge^2 V$ , not in  $\mathrm{Sym}^2 V$ . Hence the stated Hom space is zero.  $\square$

**Theorem 3.4** ( $1/\pi$  relation forces CM). *Let  $x \in \overline{\mathbb{Q}}$  satisfy the conditions of Lemma 3.1, and suppose*

$$uF_\ell(x) + v\Theta F_\ell(x) = \frac{\beta}{\pi}, \quad u, v, \beta \in \overline{\mathbb{Q}}, \quad (u, v) \neq (0, 0).$$

*Then under  $(\mathrm{GPC}_{\mathrm{tor}})$ , the corresponding elliptic curve  $E_x$  must have complex multiplication. This conclusion includes the case when  $\beta = 0$ .*

*Proof.* Take  $s_0, s_1, \delta$  from Lemma 3.1. Multiply the equality by  $(2\pi i)^2$ , and use  $1/\pi = 2i/(2\pi i)$ , obtaining  $\langle \mathrm{comp}(us_0 + vs_1), \delta \rangle - 2i\beta(2\pi i) = 0$ . The left-hand side is the sum of two matrix coefficients that come from  $M =$

$\mathrm{Sym}^2 H^1(E_x)$  and  $T = \mathbb{Q}(-1)$ . (GPC<sub>tor</sub>) says that the comparison point is Zariski dense in the motivic period torsor, then the above linear function does not only vanish at one numerical point, but also vanishes identically on the whole motivic period torsor.

Assume, for contradiction, that  $E_x$  is non-CM. By Lemma 3.3,  $M$  and  $T$  are non-isomorphic simple objects. By Lemma 3.2, their matrix coefficient spaces are direct summands and separate, so the two components must vanish separately, i.e.  $2i\beta = 0, (us_0 + vs_1) \otimes \delta = 0$ . Since  $\delta \neq 0$  and  $s_0, s_1$  are linearly independent, the second equation forces  $u = v = 0$ , a contradiction.  $\square$

**Theorem 3.5** (Relation space at CM points is at most one-dimensional). *Let  $E_x$  have CM, and set*

$$p_0 = (2\pi i)^2 F_\ell(x), \quad p_1 = (2\pi i)^2 \Theta F_\ell(x), \quad t = 2\pi i.$$

*Under (GPC<sub>tor</sub>), the  $\overline{\mathbb{Q}}$ -linear relation space among the triple  $(p_0, p_1, t)$  is at most one-dimensional.*

*Proof.* Extend the coefficient field to an algebraically closed field containing the CM field  $K$ . If  $V = H^1(E_x) = L \oplus \overline{L}$  then  $M = \mathrm{Sym}^2 V = L^{\otimes 2} \oplus (L \otimes \overline{L}) \oplus \overline{L}^{\otimes 2}$ , the polarization gives  $L \otimes \overline{L} \simeq T = \mathbb{Q}(-1)$ . The other two one-dimensional objects have Hodge types  $(2, 0)$  and  $(0, 2)$ , respectively, and are non-isomorphic to the Tate object.

The Betti tensor from Lemma 3.1 may be taken as  $\delta = \gamma^{\otimes 2}$ , where  $0 \neq \gamma \in H_1(E_x, \mathbb{Q})$ . Under the canonical decomposition  $V_B = L_B \oplus \overline{L}_B$ , the projections of  $\gamma$  to both components are nonzero; otherwise complex conjugation exchanges the two CM characters and fixes the rational vector  $\gamma$ , forcing  $\gamma = 0$ . Thus the projections of  $\delta$  to  $L^{\otimes 2}$  and  $\overline{L}^{\otimes 2}$  are also nonzero.

Suppose  $ap_0 + bp_1 + ct = 0$ . By GPC and Lemma 3.2, the two non-Tate coefficient matrix cannot be canceled by the Tate matrix coefficient. Therefore  $as_0 + bs_1 \in T_{\mathrm{dR}} \subset M_{\mathrm{dR}}$ . Conversely, whenever  $as_0 + bs_1 \in T_{\mathrm{dR}}$ , its Tate matrix coefficient determines a unique  $c$ .

Hence the relation space has the same dimension as  $W \cap T_{\mathrm{dR}}$ , where  $W = \mathrm{span}_{\overline{\mathbb{Q}}}\{s_0, s_1\}$ . By Lemma 3.1,  $\dim W = 2$ . Also  $0 \neq s_0 \in F^2 M_{\mathrm{dR}}$ , while the Tate object  $T = \mathbb{Q}(-1)$  has Hodge type  $(1, 1)$ , so  $F^2 T_{\mathrm{dR}} = 0$ .

Thus  $s_0 \notin T_{\mathrm{dR}}$ , and therefore  $\dim(W \cap T_{\mathrm{dR}}) \leq 1$ , hence the relation space is at most one-dimensional.  $\square$

**Corollary 3.6** (Unique primitive slope for fixed parameter). *Fix  $x = 1/C \neq \kappa_\ell^{-1}$ . If there are two relations*

$$A_i F_\ell(x) + B_i \Theta F_\ell(x) = \frac{\alpha_i}{\pi}, \quad i = 1, 2,$$

*then*

$$A_1 B_2 - A_2 B_1 = \det \begin{pmatrix} A_1 & B_1 \\ A_2 & B_2 \end{pmatrix} = 0.$$

*If both pairs of integers are primitive, then  $(A_2, B_2) = \pm(A_1, B_1)$ .*

*Proof.* It follows straightly from the recent theorems: If the determinant were nonzero, the two relations would be linearly independent among the three numbers  $F_\ell(x), \Theta F_\ell(x), 1/\pi$ , contradicting Theorem 3.5. In addition,

primitive integer vectors representing the same rational projective point can differ only by a sign.  $\square$

#### 4. RATIONAL HAUPTMODUL VALUES AND RING CLASS NUMBERS

**Lemma 4.1.** *Let  $x = X_\ell(\sigma) \in \mathbb{Q}$  and  $\sigma$  be a CM point. Then  $[\mathbb{Q}(j(\sigma)) : \mathbb{Q}] \leq 2$ . Moreover, when  $\ell = 1$ , the bound is 1.*

*Proof.*  $x \in \mathbb{Q}$  gives a rational point on  $Y_\ell$ . By Lemma 2.4, one can choose a preimage  $\tilde{P}$  on  $X_0(\ell)$  such that  $[\mathbb{Q}(\tilde{P}) : \mathbb{Q}] \leq 2$  and the bound is 1 when  $\ell = 1$ . The function  $j$  is defined on  $X_0(\ell)$  and defined over  $\mathbb{Q}$ , hence  $j(\sigma) = j(\tilde{P}) \in \mathbb{Q}(\tilde{P})$ .  $\square$

**Shimura reciprocity law.** For an imaginary quadratic field  $K$ , let  $\mathcal{O}_K$  be its ring of integers and take a CM point  $\tau \in \mathcal{H}$  such that  $K = \mathbb{Q}(\tau)$ . For any fractional ideal  $\mathfrak{a} \subset K$ , write  $\mathfrak{a} = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$  and define  $\tau_{\mathfrak{a}} := \omega_1/\omega_2$ ; then the assignment  $[\mathfrak{a}] \mapsto j(\tau_{\mathfrak{a}})$  gives a well-defined action of the ideal class group  $\text{Cl}(\mathcal{O}_K)$  on the set of singular moduli. In the classical case of the  $j$ -function, *Shimura reciprocity law* asserts precisely that this action is simply transitive: it acts transitively on all conjugates, meaning that for every  $\sigma \in \text{Gal}(\overline{K}/K)$  there exists an ideal class  $[\mathfrak{a}]$  such that  $\sigma(j(\tau)) = j(\tau_{\mathfrak{a}})$ ; and it acts freely, because if some class  $[\mathfrak{a}]$  fixes  $j(\tau)$ , i.e.  $j(\tau_{\mathfrak{a}}) = j(\tau)$ , then  $\mathfrak{a}$  must be principal, so the stabiliser is trivial. Consequently the number of distinct conjugates equals the class number  $h(K) = \#\text{Cl}(\mathcal{O}_K)$ , and we obtain the canonical isomorphism  $\text{Cl}(\mathcal{O}_K) \xrightarrow{\sim} \text{Gal}(H_K/K)$ , where  $H_K$  is the Hilbert class field of  $K$ ; this isomorphism is explicitly realised in CM theory by the above action of ideal classes on the  $j$ -values.

**Lemma 4.2.** *Let  $\mathcal{O}_D$  be the imaginary quadratic order corresponding to  $\sigma$ . Then  $h(D) = \#\text{Pic}(\mathcal{O}_D) = [\mathbb{Q}(j(\sigma)) : \mathbb{Q}] \leq 2$ .*

*Proof.* The ring class polynomial of discriminant  $D$  was given by

$$H_D(Y) = \prod_{[\mathfrak{a}] \in \text{Pic}(\mathcal{O}_D)} (Y - j(\mathfrak{a}))$$

lies in  $\mathbb{Z}[Y]$ . Shimura reciprocity shows that the class group acts transitively on these singular moduli; together with complex conjugation, all roots form one  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ -orbit. Therefore  $H_D$  is irreducible over  $\mathbb{Q}$  and has degree  $h(D)$ . Hence  $[\mathbb{Q}(j(\sigma)) : \mathbb{Q}] = h(D)$ . Combining with Lemma 4.1 and the result follows.  $\square$

**Lemma 4.3.** *When the integer  $f \rightarrow \infty$ , the Euler function  $\phi(f) \rightarrow \infty$ .*

*Proof.* Proof by contradiction: If there are a finite accumulation point of  $\{\phi(f)\}_{f \geq 1}$ , say  $N$ , then there are infinitely many  $k$  s.t.  $\phi(k) = N$ . Let  $a_n = \#\{k : \phi(k) = n\}$  then consider its Dirichlet generating function  $F(s)$ . By definition,

$$F(s) = \sum_{m=1}^{\infty} \frac{1}{\phi(m)^s}.$$

Since  $\phi$  is multiplicative,  $F(s)$  admits an Euler product:

$$F(s) = \prod_p \left( 1 + \sum_{k=1}^{\infty} \frac{1}{\phi(p^k)^s} \right) = \prod_p \left( 1 + \frac{1}{(p-1)^s} \frac{1}{1-p^{-s}} \right).$$

For  $\Re(s) > 1$ ,  $\frac{1}{(p-1)^s} \sim p^{-s}$ , so that the product converges absolutely; hence  $F(s)$  converges absolutely in that half-plane.

On the other hand, if some  $N$  satisfied  $a_N = \infty$ , the series would diverge for every  $s$ , while  $F(s)$  absolutely converges for every  $\Re(s) > 1$ . A contradiction!  $\square$

**Lemma 4.4** (Bounded ring class numbers give finitely many orders). *For any fixed  $H$ , there are only finitely many negative discriminants  $D$  satisfying  $h(D) \leq H$ .*

*Proof.* First assume  $D = D_K$  is a fundamental discriminant. The Siegel–Heilbronn lower bound says that for every  $\varepsilon > 0$ , there exists  $c(\varepsilon) > 0$  such that

$$h(D_K) \geq c(\varepsilon) |D_K|^{1/2-\varepsilon}.$$

Thus as the absolute value of a fundamental discriminant tends to infinity, the class number tends to infinity; there are only finitely many imaginary quadratic fields of bounded class number.

In general write  $D = f^2 D_K$ . The class number formula for orders is

$$h(f^2 D_K) = \frac{h(D_K)f}{[\mathcal{O}_K^\times : \mathcal{O}_f^\times]} \prod_{p|f} \left( 1 - \frac{(\frac{D_K}{p})}{p} \right).$$

Since

$$1 - \frac{(\frac{D_K}{p})}{p} \geq 1 - \frac{1}{p},$$

we have

$$f \prod_{p|f} \left( 1 - \frac{(\frac{D_K}{p})}{p} \right) \geq f \prod_{p|f} \left( 1 - \frac{1}{p} \right) = \varphi(f).$$

The unit index is at most 3, while by Lemma 4.3,  $\varphi(f) \rightarrow \infty$ . Thus for fixed  $D_K$ ,  $h(f^2 D_K) \rightarrow \infty$  as  $f \rightarrow \infty$ . Combining with the finite number of  $D_K$  from the previous paragraph, the result follows.  $\square$

## 5. MAIN THEOREM AND THE COMPANION BRANCH

*Proof of Theorem 1.1.* By Lemma 2.2, convergence requires  $|C| \geq \kappa_\ell$ . First treat  $C = \kappa_\ell$ . Convergence then forces  $B = 0$ , and primitivity gives  $A = \pm 1$ . Thus this boundary contributes at most one sign class.

Now assume  $C \neq \kappa_\ell$ , and set  $x = 1/C \in \mathbb{Q}$  then  $-\kappa_\ell^{-1} \leq x < \kappa_\ell^{-1}$  and  $x \neq 0$ . If

$$AF_\ell(x) + B\Theta F_\ell(x) = \frac{\alpha}{\pi},$$

then Theorem 3.4 shows that the corresponding elliptic curve  $E_x$  has CM. Let its CM order discriminant be  $D$ . Since  $x \in \mathbb{Q}$ , Lemma 4.2 gives  $h(D) \leq 2$ . By Lemma 4.4, there are only finitely many possible  $D$ .

For each fixed  $D$ , the isomorphism classes of CM elliptic curves of discriminant  $D$  are parametrised by the finite set  $\text{Pic}(\mathcal{O}_D)$ . Adding a cyclic  $\ell$ -subgroup on  $X_0(\ell)$  gives only finitely many choices. Therefore the corresponding  $Y_\ell$ -points and their Hauptmodul values  $x = X_\ell(\sigma)$  are finite in number. Among them, those satisfying  $x = 1/C$  with  $C \in \mathbb{Z}$  are certainly finite. Hence the possible  $C$ 's are finite.

For each  $C \neq \kappa_\ell$ , Corollary 3.6 shows that the primitive integer pair  $(A, B)$  has at most one sign class. Together with the already separately handled case  $C = \kappa_\ell$ , we obtain finiteness of triples  $(A, B, C)$ . Once  $(A, B, C)$  is fixed,

$$\alpha = \pi(AF_\ell(C^{-1}) + B\Theta F_\ell(C^{-1}))$$

is uniquely determined. QED!  $\square$

**Corollary 5.1** (Finite degree parameter curve version). *Let  $Y/\mathbb{Q}$  be a fixed algebraic curve equipped with two nonconstant functions defined over  $\mathbb{Q}$ ,*

$$y : Y \rightarrow \mathbf{P}^1, \quad j : Y \rightarrow \mathbf{P}^1,$$

*and suppose  $\deg(y) = d < \infty$ . Assume that for a companion branch, the parameter  $x$  comes from  $Y$ , and its Picard–Fuchs system is still the symmetric square of an elliptic  $H^1$ . If an integer denominator condition gives  $y(P) = 1/C \in \mathbb{Q}$ , then  $[\mathbb{Q}(j(P)) : \mathbb{Q}] \leq d$ . Under  $(\text{GPC}_{\text{tor}})$ , points satisfying the corresponding  $1/\pi$  relation must be CM points, so  $h(D) \leq d$ , and therefore primitive integer solutions still form a finite set.*

*Proof.* The rational number  $y(P)$  gives a rational point on  $\mathbb{P}^1$ . The residue field degree of any closed point preimage under the finite map  $y$  is at most  $d$ , hence  $[\mathbb{Q}(P) : \mathbb{Q}] \leq d$ . Since  $j$  is defined over  $\mathbb{Q}$ ,  $j(P) \in \mathbb{Q}(P)$ , so  $[\mathbb{Q}(j(P)) : \mathbb{Q}] \leq d$ . The rest of the argument is the same as in the main theorem: the period relation forces CM, CM theory identifies the  $j$ -degree with the ring class number, and bounded ring class numbers give finitely many discriminants and finitely many parameter points.  $\square$

## 6. COMPLETE CLASSIFICATION

The first five sections only prove finiteness. In this section we add the explicit enumeration of low-class-number CM points to the argument, thereby obtaining the complete primitive integer tables for the standard  $1A, 2A, 3A, 4A$  branches and its Euler-companion branches.

**Standard A-branch.** Campbell–Cooper–Ye’s Theorem 2.4 gives a uniform formula [2]: if  $X = 1/C$  and the projective slope  $\lambda = A/B$  are both rational, then on the corresponding positive and negative nome branches,

(\*)

$$\sum_{n \geq 0} T_\ell(n)(n+\lambda)X^n = \frac{1}{2\pi\sqrt{1-\kappa_\ell X}} \begin{cases} \sqrt{\ell/N}, & \text{positive nome,} \\ \sqrt{L_\ell/N}, & \text{negative nome,} \end{cases} \quad L_\ell = \frac{4\ell}{\gcd(4, \ell)}.$$

Chen–Glebov–Goeanka show that the corresponding choice of rational uniformizer is completely enumerated by the list of imaginary quadratic orders of class number  $\leq 2$ , and that their Clausen single-sum cases agree exactly with the rational Ramanujan-type single-sum tables for levels 1–4 [3,

Sec. 10]. This arithmetic enumeration is the input that upgrades “finite” to “complete list” in this subsection.

**Proposition 6.1** (From  $(X, \lambda, N)$  to primitive integer quadruples). *Let  $X = 1/C \in \mathbf{Q}$ ,  $C \in \mathbf{Z} \setminus \{0\}$ , and write*

$$\lambda = \frac{A}{B}, \quad A, B \in \mathbf{Z}, \quad B > 0, \quad \gcd(A, B) = 1.$$

*If the series corresponding to (7.1) converges, then*

$$\sum_{n \geq 0} T_\ell(n) \frac{A + Bn}{C^n} = \frac{\alpha}{\pi},$$

*where*

$$(\clubsuit) \quad \alpha = \frac{B}{2\sqrt{1 - \kappa_\ell/C}} \begin{cases} \sqrt{\ell/N}, & \text{positive nome,} \\ \sqrt{L_\ell/N}, & \text{negative nome.} \end{cases}$$

*In particular, the  $\alpha$  in the tables automatically lies in an extension of degree at most two.*

*Proof.* Multiply  $(*)$  by  $B$ , and use  $B(n + A/B) = A + Bn$ . Since  $C, N, \ell, L_\ell \in \mathbf{Q}$ , the right-hand side contains only one square root, so  $[\mathbf{Q}(\alpha) : \mathbf{Q}] \leq 2$ .  $\square$

**Lemma 6.2.** *Under  $(\text{GPC}_{\text{tor}})$ , the standard  $A$  branch at  $C = \kappa_\ell$  yields no primitive integer  $1/\pi$  solutions.*

*Proof.* By Lemma 2.2, at  $C = \kappa_\ell$  convergence forces  $B = 0$ , and primitivity further gives  $A = \pm 1$ . The point  $x = \kappa_\ell^{-1}$  is a finite elliptic branch point of the modular curve. After passing to a finite cover that removes orbifold automorphisms, the weight-two Hodge section still gives a nonzero vector  $0 \neq s_0 \in F^2 \text{Sym}^2 H_{\text{dR}}^1(E_x)$ . If  $F_\ell(x) = \alpha/\pi$ , then the same period torsor argument as in Theorem 3.4 yields an identity between this matrix coefficient of  $\text{Sym}^2 H^1(E_x)$  and a Tate matrix coefficient. In the CM decomposition, the non-Tate matrix coefficients must vanish separately, so  $s_0$  would have to lie in the Tate line  $\mathbf{Q}(-1)_{\text{dR}}$ . However,  $F^2 \mathbf{Q}(-1)_{\text{dR}} = 0$ , contradicting  $s_0 \neq 0$ . done.  $\square$

**Theorem 6.3** (Complete primitive integer classification of standard  $A$  branches). *Assume  $(\text{GPC}_{\text{tor}})$  holds for the elliptic curves appearing in the proof. Up to the overall sign*

$$(A, B, \alpha) \sim (-A, -B, -\alpha),$$

*the convergent primitive integer solutions of the standard  $1A, 2A, 3A, 4A$  branches are exactly the 36 entries listed in Tables 1–4:*

$$11 + 12 + 9 + 4 = 36.$$

*Except for the Bauer formula  $(A, B, C) = (1, 4, -64)$  in  $4A$ , all other 35 entries are absolutely convergent.*

*Proof.* Take any convergent primitive solution. If  $C = \kappa_\ell$ , it is excluded by Lemma 6.2. Otherwise set  $x = 1/C$ . Theorem 3.4 forces  $E_x$  to have CM; since  $x \in \mathbf{Q}$ , Lemma 4.2 gives  $h(D) \leq 2$ .

Chen–Glebov–Goeanka give a complete enumeration of rational uniformizers

for these low-class-number imaginary quadratic orders [3, Sec. 10]; Campbell–Cooper–Ye’s tables for levels 1–4 [2, Tables 5–11] give the corresponding rational parameters  $X$  and slopes  $\lambda$ . Applying simultaneously the conditions  $X = 1/C$ ,  $C \in \mathbf{Z}$ ,  $\lambda \in \mathbf{Q}$ , and the convergence conditions of Lemma 2.2, yields exactly the following 36 rows. There are two additional rational CM candidates

$$(\spadesuit) \quad (\ell; A, B, C) = (2; 1, 5, -144), (3; 4, 15, -27),$$

but they satisfy respectively  $|C| < 256$  and  $|C| < 108$ , so the original  $A$ -series diverges and they are not counted.

For each surviving  $C$ , Corollary 3.6 says that the primitive slope is unique up to the overall sign; and Proposition 6.1 gives the nonzero  $\alpha$  in the table. Hence there are neither omissions nor repetitions. Finally, the convergence type is read off directly from Lemma 2.2; only  $C = -\kappa_4 = -64$  with  $B \neq 0$ , the Bauer case, is conditionally convergent.  $\square$

TABLE 1. All primitive integer solutions for Level 1A (taking  $B > 0$ )

| nome | $N$ | $A$      | $B$       | $C$                 | $\alpha$             |
|------|-----|----------|-----------|---------------------|----------------------|
| +    | 2   | 3        | 28        | 8000                | $5\sqrt{5}$          |
| +    | 3   | 1        | 11        | 54000               | $5\sqrt{15}/6$       |
| +    | 4   | 5        | 63        | 287496              | $11\sqrt{33}/4$      |
| +    | 7   | 8        | 133       | 16581375            | $85\sqrt{255}/54$    |
| -    | 7   | 8        | 63        | -3375               | $5\sqrt{15}$         |
| -    | 11  | 15       | 154       | -32768              | $32\sqrt{2}$         |
| -    | 19  | 25       | 342       | -884736             | $32\sqrt{6}$         |
| -    | 27  | 31       | 506       | -12288000           | $160\sqrt{30}/9$     |
| -    | 43  | 263      | 5418      | -884736000          | $640\sqrt{15}/3$     |
| -    | 67  | 10177    | 261702    | -147197952000       | $1760\sqrt{330}$     |
| -    | 163 | 13591409 | 545140134 | -262537412640768000 | $426880\sqrt{10005}$ |

**Euler-companion branch.** In this subsection we fix the meaning of “companion” to avoid the ambiguity caused by the not entirely consistent notation “B” and “C” branches in the literature. Set

$$(1) \quad (K_1, K_2, K_3, K_4) = (432, 64, 27, 16) = \frac{1}{4}(\kappa_1, \kappa_2, \kappa_3, \kappa_4).$$

For  $\ell = 1, 2, 3, 4$ , define the Euler-companion coefficients

$$(2) \quad U_\ell(n) = \sum_{k=0}^n T_\ell(k) \binom{n+k}{n-k} (-K_\ell)^{n-k}.$$

In this paper we denote these four sequences respectively by  $1B, 2B, 3B, 4C$ .

**Remark 6.4.** In the notation of Chen–Glebov–Goenka, the last branch is called  $2C$ ; we use  $4C$  to remain consistent with the level 4A single-sum sequence that generates it.

TABLE 2. All primitive integer solutions for Level 2A (taking  $B > 0$ )

| nome | $N$ | $A$  | $B$   | $C$         | $\alpha$         |
|------|-----|------|-------|-------------|------------------|
| +    | 2   | 1    | 7     | 648         | $9/2$            |
| +    | 3   | 1    | 8     | 2304        | $2\sqrt{3}$      |
| +    | 5   | 1    | 10    | 20736       | $9\sqrt{2}/4$    |
| +    | 9   | 3    | 40    | 614656      | $49\sqrt{3}/9$   |
| +    | 11  | 19   | 280   | 2509056     | $18\sqrt{11}$    |
| +    | 29  | 1103 | 26390 | 24591257856 | $9801\sqrt{2}/4$ |
| -    | 5   | 3    | 20    | -1024       | 8                |
| -    | 7   | 8    | 65    | -3969       | $9\sqrt{7}$      |
| -    | 9   | 3    | 28    | -12288      | $16\sqrt{3}/3$   |
| -    | 13  | 23   | 260   | -82944      | 72               |
| -    | 25  | 41   | 644   | -6635520    | $288\sqrt{5}/5$  |
| -    | 37  | 1123 | 21460 | -199148544  | 3528             |

TABLE 3. All primitive integer solutions for Level 3A (taking  $B > 0$ )

| nome | $N$ | $A$ | $B$   | $C$       | $\alpha$        |
|------|-----|-----|-------|-----------|-----------------|
| +    | 2   | 1   | 6     | 216       | $3\sqrt{3}$     |
| +    | 4   | 2   | 15    | 1458      | $27/4$          |
| +    | 5   | 4   | 33    | 3375      | $15\sqrt{3}/2$  |
| -    | 9   | 1   | 5     | -192      | $4\sqrt{3}/3$   |
| -    | 17  | 7   | 51    | -1728     | $12\sqrt{3}$    |
| -    | 25  | 1   | 9     | -8640     | $4\sqrt{15}/5$  |
| -    | 41  | 53  | 615   | -110592   | $96\sqrt{3}$    |
| -    | 49  | 13  | 165   | -326592   | $108\sqrt{7}/7$ |
| -    | 89  | 827 | 14151 | -27000000 | $1500\sqrt{3}$  |

TABLE 4. All primitive integer solutions for Level 4A (taking  $B > 0$ )

| nome | $N$ | $A$ | $B$ | $C$  | $\alpha$    | Convergence |
|------|-----|-----|-----|------|-------------|-------------|
| +    | 3   | 1   | 6   | 256  | 4           | absolute    |
| +    | 7   | 5   | 42  | 4096 | 16          | absolute    |
| -    | 2   | 1   | 4   | -64  | 2           | conditional |
| -    | 4   | 1   | 6   | -512 | $2\sqrt{2}$ | absolute    |

**Lemma 6.5** (Euler generating function transform). *Let*

$$\tilde{\mathfrak{F}}_\ell(Y) = \sum_{n \geq 0} U_\ell(n) Y^n.$$

*Then in a neighbourhood of  $Y = 0$  we have identically*

$$(3) \quad \tilde{\mathfrak{F}}_\ell(Y) = \frac{1}{1 + K_\ell Y} \tilde{\mathfrak{F}}_\ell\left(\frac{Y}{(1 + K_\ell Y)^2}\right).$$

*Proof.* From (2), put  $n = k + j$  and interchange summation gives

$$\begin{aligned}\tilde{\mathfrak{F}}_\ell(Y) &= \sum_{k \geq 0} T_\ell(k) Y^k \sum_{j \geq 0} \binom{2k+j}{j} (-K_\ell Y)^j \\ &= \sum_{k \geq 0} T_\ell(k) Y^k (1 + K_\ell Y)^{-2k-1} \\ &= \frac{1}{1 + K_\ell Y} \sum_{k \geq 0} T_\ell(k) \left( \frac{Y}{(1 + K_\ell Y)^2} \right)^k.\end{aligned}$$

The last sum is precisely  $\mathfrak{F}_\ell$ . □

**Lemma 6.6** (Explicit transformations). *Let*

$$Y = \frac{1}{c}, \quad c \in \mathbf{Z}, \quad c \neq -K_\ell,$$

and define

$$(4) \quad x = \frac{Y}{(1 + K_\ell Y)^2} = \frac{1}{J}, \quad J = \frac{(c + K_\ell)^2}{c}.$$

If

$$\tilde{\mathfrak{G}}_\ell(A_c, B_c, c) = \sum_{n \geq 0} U_\ell(n) \frac{A_c + B_c n}{c^n},$$

then

$$(5) \quad \tilde{\mathfrak{G}}_\ell(A_c, B_c, c) = \frac{1}{J} (a F_\ell(J^{-1}) + b \Theta F_\ell(J^{-1})),$$

where

$$(6) \quad a = A_c(c + K_\ell) - B_c K_\ell, \quad b = B_c(c - K_\ell).$$

If  $B_c \neq 0$  and  $c \neq K_\ell$ , write  $\lambda_c = A_c/B_c$  and  $\lambda_A = a/b$ . Then

$$(7) \quad \lambda_A = \frac{(c + K_\ell)\lambda_c - K_\ell}{c - K_\ell}, \quad \lambda_c = \frac{K_\ell + (c - K_\ell)\lambda_A}{c + K_\ell}.$$

If  $(a, b) = g(A, B)$ , where  $\gcd(A, B) = 1$ , and

$$A \mathfrak{F}_\ell(J^{-1}) + B \Theta \mathfrak{F}_\ell(J^{-1}) = \frac{\alpha_A}{\pi},$$

then

$$(8) \quad \alpha_c = \frac{g}{J} \alpha_A.$$

*Proof.* Apply  $\Theta_Y = Y d/dY$  to (3). If  $r = 1 + K_\ell Y$ , then

$$Y \frac{d}{dY} \log \frac{Y}{r^2} = \frac{1 - K_\ell Y}{1 + K_\ell Y}.$$

Therefore

$$\Theta_Y \tilde{\mathfrak{F}}_\ell(Y) = -\frac{K_\ell Y}{r^2} \tilde{\mathfrak{F}}_\ell(x) + \frac{1 - K_\ell Y}{r^2} \Theta \tilde{\mathfrak{F}}_\ell(x).$$

Substituting  $Y = 1/c$  into  $A_c \tilde{\mathfrak{F}}_\ell + B_c \Theta_Y \tilde{\mathfrak{F}}_\ell$  and simplifying gives (5)–(6). Equation (7) is just the simplification of  $a/b$  and its inverse; if  $(a, b) = g(A, B)$ , then (5) immediately gives (8). □

**Corollary 6.7** (Discriminant condition for integer denominators). *If  $J \in \mathbb{Z}$ , then (4) is equivalent to*

$$(9) \quad c^2 + (2K_\ell - J)c + K_\ell^2 = 0.$$

*Hence a necessary condition for an integer  $c$  to exist is*

$$(10) \quad \Delta = J(J - 4K_\ell) \text{ is an integer square.}$$

*Conversely, if  $\Delta$  is a square and  $(J - 2K_\ell \pm \sqrt{\Delta})/2 \in \mathbb{Z}$ , then the two integer roots are*

$$(11) \quad c = \frac{J - 2K_\ell \pm \sqrt{J(J - 4K_\ell)}}{2}.$$

**Lemma 6.8** (Convergence boundary for the companion). *For  $U_\ell(n)$  there exists a constant  $d_\ell > 0$  such that the leading asymptotics have sign  $(-1)^n$ , and*

$$(12) \quad U_\ell(n) = (-1)^n K_\ell^n \times \begin{cases} d_1 n^{-1/3} (1 + o(1)), & \ell = 1, \\ d_2 n^{-1/2} (1 + o(1)), & \ell = 2, \\ d_3 n^{-2/3} (1 + o(1)), & \ell = 3, \\ d_4 \frac{\log n}{n} (1 + o(1)), & \ell = 4. \end{cases}$$

*Therefore*

$$(13) \quad \begin{array}{l|l} |c| > K_\ell & \text{absolutely convergent,} \\ c = K_\ell & \text{convergent iff } B_c = 0, \text{ and only conditionally,} \\ c = -K_\ell & \text{divergent,} \\ |c| < K_\ell & \text{divergent.} \end{array}$$

*Proof.* From (3), the dominant singularity closest to the origin is  $Y = -K_\ell^{-1}$ . Put  $r = 1 + K_\ell Y$ . As  $r \rightarrow 0$ ,

$$\frac{Y}{r^2} \longrightarrow -\infty.$$

Using the standard connection formula for

$${}_3F_2 \left( a_\ell, \frac{1}{2}, 1 - a_\ell; 1, 1; \kappa_\ell x \right)$$

as  $x \rightarrow -\infty$ : when  $a_\ell = 1/6, 1/4, 1/3$ , the minimal parameter term gives respectively singularities of order  $r^{-2/3}, r^{-1/2}, r^{-1/3}$ ; when  $a_4 = 1/2$  there is a triple coincidence, and the dominant singularity is  $(\log r)^2$ . The Darboux–Flajolet transfer theorem then gives (12).

When  $|c| > K_\ell$ , the root test gives absolute convergence; when  $|c| < K_\ell$ , the general term does not tend to zero. At  $c = K_\ell$ , the factor  $(-1)^n$  remains; the  $A_c$ -part converges by the alternating test, while if  $B_c \neq 0$ , multiplying by  $n$  prevents the general term from tending to zero. At  $c = -K_\ell$ , the alternating sign is cancelled; the  $A_c$ -part behaves respectively as  $n^{-1/3}, n^{-1/2}, n^{-2/3}, (\log n)/n$ , all divergent; if  $B_c \neq 0$  convergence is even more impossible.  $\square$

**Theorem 6.9** (Complete primitive integer classification of Euler–companion branches). *Under  $(\text{GPC}_{\text{tor}})$ , in the Euler–companion branches  $1B, 2B, 3B, 4C$*

as defined in (2), up to the overall sign there are exactly 5 convergent primitive integer solutions, all listed in Table 5. Among them 1B has no solution, 2B has three, 3B and 4C each have one; all five are absolutely convergent.

*Proof.* First, by Lemma 6.8,  $|c| < K_\ell$  and  $c = -K_\ell$  are all excluded. If  $c = K_\ell$ , convergence forces  $B_c = 0$ . From (3), then

$$\tilde{\mathfrak{F}}_\ell(K_\ell^{-1}) = \frac{1}{2}\tilde{\mathfrak{F}}_\ell(\kappa_\ell^{-1}),$$

so if it were an algebraic number times  $1/\pi$ , this would contradict Lemma 6.2. Thus we only need consider  $|c| > K_\ell$ .

For such a solution, (4) gives a rational A-side modular parameter  $J$ , and (5) converts the companion relation into an A-side linear period relation in the same Picard–Fuchs system; hence GPC again forces the corresponding point to be CM. Chen–Glebov–Goenka give a complete enumeration of low-class-number rational CM uniformizers for Euler double-sum branches; their Table 15 gives all rational 2B, 3B Euler cases, and their Table 13 together with equation (49) gives the rational 2C (our 4C) case [3, Tables 13,15]. Combining with (4), after excluding the generic exceptional values  $J = 0$  (corresponding to  $c = -K_\ell$ ) and  $J = 4K_\ell$  (corresponding to  $c = K_\ell$ ), the non-boundary CM parameters that can produce integer  $c$  are exactly

|      | Branch | $J$   | Integer roots $c$ from (11) |
|------|--------|-------|-----------------------------|
| (14) | 1B     | none  | none                        |
|      | 2B     | 648   | 8, 512                      |
|      | 2B     | −144  | −16, −256                   |
|      | 2B     | −3969 | −1, −4096                   |
|      | 3B     | −192  | −3, −243                    |
|      | 4C     | −8    | −8, −32.                    |

Lemma 6.8 retains only

$$(15) \quad 512, -256, -4096, -243, -32.$$

The corresponding primitive slopes and right-hand constants are obtained from the same two tables of Chen–Glebov–Goenka; alternatively, in the first three and the fourth cases they may be checked consistently from the corresponding A-side slopes using Lemma 6.6. This yields the five rows of Table 5. Since each row satisfies  $|c| > K_\ell$ , all five series are absolutely convergent. The complete CM enumeration together with the one-dimensionality of the relation space for fixed parameters excludes any other primitive slopes.  $\square$

The corresponding five identities can be written uniformly as

$$(16) \quad \sum_{n=0}^{\infty} U_\ell(n) \frac{A + Bn}{C^n} = \frac{\alpha}{\pi}.$$

In particular, level 4C can also be written in a pure convolution form. Indeed,

$$(17) \quad U_4(n) = (-1)^n \sum_{k=0}^n \binom{2k}{k}^2 \binom{2n-2k}{n-k}^2,$$

TABLE 5. All Euler–companion primitive integer solutions  
(taking  $B > 0$ )

| Branch | $A$                                      | $B$ | $C$   | $\alpha$        | Reference               |
|--------|------------------------------------------|-----|-------|-----------------|-------------------------|
| $1B$   | no convergent primitive integer solution |     |       |                 |                         |
| $2B$   | 2                                        | 9   | 512   | 4               | [3, Table 15]           |
| $2B$   | 0                                        | 1   | -256  | $4\sqrt{3}/9$   | [3, Table 15]           |
| $2B$   | 1                                        | 9   | -4096 | $64\sqrt{7}/49$ | [3, Table 15]           |
| $3B$   | 1                                        | 8   | -243  | $3\sqrt{3}$     | [3, Table 15]           |
| $4C$   | 0                                        | 1   | -32   | 2               | [3, Table 13, Eq. (49)] |

so the last row is equivalent to

$$(18) \quad \sum_{n=0}^{\infty} \frac{n}{32^n} \sum_{k=0}^n \binom{2k}{k}^2 \binom{2n-2k}{n-k}^2 = \frac{2}{\pi}.$$

**Corollary 6.10** (Total classification for levels 1–4). *Under the GPC assumption and the branch definitions in this paper, the convergent primitive integer  $1/\pi$  series in the standard  $A$  branches and the Euler–companion branches have, up to the overall sign, exactly*

$$36 + 5 = 41$$

*symbol classes, with branch counts*

$$(19) \quad \begin{array}{c|cccccccc} \text{Branch} & 1A & 1B & 2A & 2B & 3A & 3B & 4A & 4C \\ \hline \text{Count} & 11 & 0 & 12 & 3 & 9 & 1 & 4 & 1. \end{array}$$

*If  $(A, B, C)$  and  $(-A, -B, C)$  are regarded as distinct integer triples, then there are 82 triples. For all 41 representatives,  $\alpha$  satisfies  $[\mathbf{Q}(\alpha) : \mathbf{Q}] \leq 2$ , but this degree bound is not an assumption of the GPC finiteness argument.*

#### REFERENCES

- [1] J.-B. Bost and F. Charles, *Some remarks concerning the Grothendieck period conjecture*, J. Reine Angew. Math. 714 (2016), 175–208; arXiv:1307.1045.
- [2] J. M. Campbell, S. Cooper and D. Ye, *Quadratic irrational analogues of Ramanujan’s series for  $1/\pi$* , arXiv:2602.09352 (2026).
- [3] I. Chen, G. Glebov and R. Goenka, *Chudnovsky–Ramanujan type formulae for non-compact arithmetic triangle groups*, J. Number Theory 241 (2022), 603–654; arXiv:1909.07350.
- [4] Y. André, *Une introduction aux motifs*, Panoramas et Synthèses 17, Société Mathématique de France, 2004.
- [5] D. A. Cox, *Primes of the Form  $x^2 + ny^2$* , 2nd ed., Wiley, 2013.
- [6] N. Katz,  *$p$ -adic properties of modular schemes and modular forms*, in Modular Functions of One Variable III, Lecture Notes in Mathematics 350, Springer, 1973.